Weather Big Data Analytics

Frontend Handover and API Contract | Version 1.0 | 19 Oct 2026

For the React frontend engineer. Source of truth is the backend PRD; the live OpenAPI/Swagger UI at **/docs** supersedes this document once the backend is running. **Status:** this is the agreed specification. The backend is built in phases, so endpoints become available progressively; build against this contract with a mock layer (for example MSW) until each endpoint lands, then switch to the real API.

Part 1. What the Frontend Must Do

1.1 Ground rules

- Talk **only** to the backend REST API (base URL from `VITE_API_BASE_URL`, local default `http://localhost:8000`). Never connect to PostgreSQL or to Open-Meteo, data.gov.in or RSS feeds directly.
- All timestamps from the API are **UTC** (ISO 8601 with Z). Display in the user's local time zone and offer a UTC toggle on data tables.
- A measurement that is `null` means **not provided**. Show an en dash, never 0. Charts must show gaps, not zeros.
- Always show **source attribution and trust badge** next to data (Section 1.4 wording rules are mandatory).
- Treat every string from the API as plain text. Never render it as HTML. Never auto-embed or auto-fetch evidence or news URLs.
- Dev origin `http://localhost:5173` is allowed by CORS. Auth is a Bearer header; no cookies.

1.2 Screens and UI features to implement

Public (no login)

Screen	UI features	Endpoints
Dashboard	Period selector (24h / 7d / 30d) and location filter. KPI cards: record counts, locations with data, temperature min/mean/max, wind mean speed and max gust, verified report count. Latest-weather card per location (temperature, humidity, rain, wind, 'as of' time, basis badge). Source freshness panel (fresh / stale / never / disabled). Handle `missing_location_ids` with a 'no data yet' card.	analytics/summary, weather/latest, locations
Weather explorer	Filters: location, kind toggle (observation / forecast), date range (max 31 days), source. Sortable paginated table; row opens detail drawer. Kind, basis and trust badges on every row. Attribution footer from `source.attribution_text`.	weather, weather/{id}, locations
Trends	Controls: metric, location, kind, date range (max 366 days), interval (hour/day/week/month), aggregation. Line chart for temperature, humidity, wind; bar chart for rainfall (sum). Gaps for null buckets; tooltip shows value, unit and `sample_count`. Optional trend-line toggle using `trend.slope_per_day`. Block requests that would exceed 2000 buckets (server returns 422).	analytics/timeseries
Location comparison	Sortable table or cards: latest observation time, temperature range, total rainfall, wind, record count, data coverage bar, verified report count.	analytics/locations
Reports map and list	Map plus list of **verified** reports. Filters: event type, location, date range. Marker detail panel with severity, description, time, 'Verified by reviewer' badge. Evidence shows type and description only (`has_link` flag; no URL for public).	reports, reports/{id}, locations
Submit report	Form: event type, severity (optional), description, location name (optional), map pin or latitude/longitude (optional 'use my location'), event time, evidence entries (max 5), contact email (optional, explain it stays private). Client validation mirrors Section 2.7. On 201 show the **tracking token once** with a copy button and a clear warning that it cannot be shown again. Handle 422 per field, 429 (show Retry-After), 409 duplicate.	POST reports
Track my report	Enter report ID and tracking token; send token in `X-Tracking-Token` header; show status (pending / verified / rejected) and reviewed time. Never show notes. 404 means 'not found or token invalid'.	reports/{id}
Events analytics	Group-by selector (event type, location, day, status) with bar or line charts and totals. Public view contains verified data only.	analytics/events
News feed	List with source, category, published time, link out (new tab, rel='noopener noreferrer'). Fixed label: 'News item, not a verified observation'.	news
Anomalies (if enabled)	Table per location and metric: time, value, baseline median, score, plain-language explanation. Persistent banner: 'Statistical flag only. Not an official warning.' Neutral informational styling; never red alert styling or words like warning, alert, emergency. Handle `insufficient_data` state.	analytics/anomalies

Screen	UI features	Endpoints
About and data sources	Trust-class legend, per-source attribution and license links from the API, statement that Open-Meteo data is model-derived (not station readings and not official).	analytics/summary (sources)

Staff (reviewer and admin, login required)

Screen	UI features	Endpoints	Role
Login	Email and password form. Same generic error for every failure (never reveal which part was wrong). Show 429 cool-down. Keep token in memory (optionally sessionStorage). On any 401, clear session and redirect to login. Show session expiry (expires_in).	auth/login, auth/me	any
Moderation queue	Pending reports, oldest first. Filters: event type, location, date range. Shows contact email (private data: do not log or copy to analytics tools). Evidence links open in a new tab with rel='noopener noreferrer', never embedded.	admin/reports/pending	reviewer, admin
Report review	Report detail plus review history timeline (reviewer, from/to status, reason code, private notes, time). Action buttons depend on role and status: **Verify** and **Reject** (reviewer, admin; reject needs notes or reason code), **Retract verified** and **Reopen rejected** (admin only, notes required). Always send `expected_row_version`. On 409 `stale_version` reload and tell the user someone else changed it; on `invalid_transition` refresh the allowed actions.	admin/reports/{id}/reviews, POST admin/reports/{id}/review	reviewer, admin
Staff analytics	When logged in, send the token on analytics calls so pending and rejected counts appear in summary, locations and events. Render those fields only if present.	analytics/*	reviewer, admin
Ingestion monitor	Runs table (filters: source, status, date) with status badges and counts (fetched, inserted, updated, skipped, failed). Run detail with sanitized errors. 'Run now' per source returns 202; poll the run until finished. Handle 409 `already_running` and `source_not_configured`. Explain that runs stay queued if the worker is not running.	admin/ingestion/runs, runs/{id}, POST admin/ingestion/{source}/run	admin
Sources	List with freshness, last success, configured flag, enable toggle, interval (300 to 86400 s), display name, attribution, JSON config editor. Show server validation errors per field. Enabling an unconfigured source returns 409.	admin/sources, PATCH admin/sources/{id}	admin

1.3 Cross-cutting requirements

- **Every data view needs four states:** loading, empty, error (with request_id shown in small text for support) and success.
- **One API client module** that adds the Bearer token, parses the error envelope, maps 401 to logout, 429 to a cool-down using Retry-After, and 422 `details[].field` to form fields.
- **Pagination UI** on every list (limit default 50, max 200; weather max 500) using `total`, `limit`, `offset`.
- **Role-based UI:** hide controls the role cannot use, but never rely on hiding; the server enforces permissions and the UI must handle 403.
- **Debounce** filter inputs; cancel stale requests; do not poll faster than every 5 seconds.
- **Responsive and accessible:** usable on mobile, keyboard navigable, charts with text alternatives, colour not the only signal for status.
- **Config:** keep the API base URL in environment config. Do not hardcode data, tokens or secrets.

1.4 Mandatory wording rules (trust model)

Data	Required label	Never say
Open-Meteo records (trust_level provider_model)	'Model-derived observation' or 'Forecast'	Official, station reading, measured
data.gov.in records (trust_level official)	'Official data' with attribution from API	Anything not backed by the API trust_level
News items	'News item, not verified'	Observation, verified, alert
Citizen reports	'Pending review' / 'Verified by reviewer' / 'Rejected'	Official, confirmed by authorities
Anomalies	'Statistical flag only. Not an official warning.'	Warning, alert, emergency

Part 2. API Contract

2.1 Conventions

Item	Rule
Base URL	`{API_BASE}/api/v1`. Health endpoints at root: `/health/live`, `/health/ready`. Interactive docs: `/docs`.
Format	JSON only, snake_case. UUIDs and timestamps are strings; integer ids are numbers.
Auth	`Authorization: Bearer <access_token>` from login (30 min expiry, no refresh token; log in again on 401).
Time filters	`from` inclusive, `to` exclusive, ISO 8601 with timezone (naive times rejected). Max span 31 days for record lists, 366 days for aggregates.
Pagination	Query `limit` (default 50, max 200; weather 500) and `offset`. Response `{items, total, limit, offset}`.
Sorting	`sort=field` or `sort=-field` (descending), allow-listed per endpoint.
CORS	Allowed origin: the configured frontend origin (dev: http://localhost:5173). Methods GET, POST, PATCH, OPTIONS. Headers Authorization, Content-Type, X-Tracking-Token, X-Request-ID. No credentials or cookies.
Request id	Every response carries `X-Request-ID`; errors include it as `request_id`.
Rate limits	Login 5/min, report submit 5/hour, tracking lookup 20/min, public reads 120/min (per IP, configurable). Exceeding returns 429 with Retry-After.

2.2 Error envelope and status codes

```
{ "error": { "code": "validation_error", "message": "Request validation failed",
  "details": [ { "field": "latitude", "issue": "must be between -90 and 90" } ],
  "request_id": "b1f0c2d4" } }
```

HTTP	code	Frontend handling
400	bad_request	Generic error message
401	unauthorized	Clear session, go to login (on login screen show generic failure)
403	forbidden	Show 'not permitted'; hide the action
404	not_found	Not found page. Also returned for non-public reports
409	conflict, invalid_transition, stale_version, already_running, source_not_configured	Show message; reload data for stale_version / invalid_transition
413	payload_too_large	Shorten input
422	validation_error	Map details[].field to form fields
429	rate_limited	Disable action until Retry-After elapses
500 / 503	internal_error / service_unavailable	Retry later; show request_id

2.3 Enumerations

Name	Values
event_type	heavy_rain, flood, heatwave, cold_wave, storm, cyclone, drought, landslide, lightning, hail, fog, other
severity	low, moderate, high, extreme (optional)
report status	pending, verified, rejected
review decision	verify, reject, reopen
weather kind	observation, forecast (all, for list filter only)
measurement_basis	station, model_analysis, model_forecast, unknown
trust_level	official, provider_model, news, citizen
source freshness	fresh, stale, never, disabled
timeseries metric	temperature_c, relative_humidity_pct, precipitation_mm, wind_speed_kmh, wind_gust_kmh
interval / agg	hour, day, week, month / avg, min, max, sum (precipitation only), count
ingestion status	queued, running, succeeded, partial, failed, abandoned
roles	reviewer, admin
units	degrees Celsius, percent, millimetres, km/h, degrees (0 to 360)

2.4 Endpoint reference

Public

Endpoint	Query / body	Success	Notes
GET /health/live, /health/ready	none	200 {status}	Ready returns 503 with per-check status if database is down
GET /locations	none	200 {items:[{id, slug, name, region, country_code, latitude, longitude}]}	Active locations; use for all selectors
GET /weather	location_id, kind (default observation), source, from, to, sort (valid_at / -valid_at), limit, offset	200 paginated WeatherRecord	Default window: 24 h back (observation) or next 72 h (forecast)
GET /weather/latest	location_id (optional), kind	200 {as_of, items:[WeatherRecord], missing_location_ids}	Newest record per location
GET /weather/{id}	integer id	200 WeatherRecord	404 if absent
GET /news	source, category, from, to, limit, offset	200 paginated NewsItem	category: news, advisory_feed
POST /reports	Body in 2.7	201 {id, status:'pending', submitted_at, tracking_token}	Token shown once. Sending a status field gives 422
GET /reports	event_type, location_id, from, to (on event_time), sort, limit, offset. Staff only: status = pending / verified / rejected / all	200 paginated PublicReport (verified only for anonymous)	status filter without staff token gives 403
GET /reports/{id}	UUID. Optional header X-Tracking-Token	200 PublicReport	Non-verified visible only to staff or matching token; else 404
GET /analytics/summary	from, to (default 7 days), location_id	200 Summary	More fields with staff token
GET /analytics/timeseries	metric, location_id, from, to (required), interval, agg, kind, fill_gaps, include_trend	200 Timeseries	Max 2000 buckets
GET /analytics/locations	from, to, sort	200 {items:[LocationStats]}	Per active location
GET /analytics/events	from, to, event_type, location_id, group_by, status (staff)	200 {group_by, items:[{key, count}], totals}	Public: verified only
GET /analytics/anomalies	location_id, metric, from, to, baseline_days, threshold	200 {items:[Anomaly], status}	May return status insufficient_data

Authentication and staff

Endpoint	Body / query	Success	Role
POST /auth/login	{email, password}	200 {access_token, token_type:'bearer', expires_in, user:{id, email, display_name, role}}	public
GET /auth/me	none	200 user object	any staff
GET /admin/reports/pending	event_type, location_id, from, to, limit, offset (oldest first)	200 paginated StaffReport	reviewer, admin
GET /admin/reports/{id}/reviews	none	200 {items:[Review]}	reviewer, admin
POST /admin/reports/{id}/review	{decision, notes?, reason_code?, expected_row_version}	200 StaffReport	reviewer, admin (retract and reopen: admin)
GET /admin/ingestion/runs	source, status, from, to, limit, offset	200 paginated Run	admin
GET /admin/ingestion/runs/{run_id}	none	200 Run with details	admin
POST /admin/ingestion/{source_name}/run	none	202 {run_id, status:'queued'}	admin
GET /admin/sources	none	200 {items:[Source]}	admin
PATCH /admin/sources/{source_id}	any of is_enabled, interval_seconds, display_name, attribution_text, config	200 Source	admin

2.5 Allowed review transitions

From	Decision	To	Role	Notes
pending	verify	verified	reviewer, admin	optional
pending	reject	rejected	reviewer, admin	required (notes or reason_code)
verified	reject	rejected	admin	required
rejected	reopen	pending	admin	required

2.6 Response schemas (examples)

WeatherRecord

```
{ "id": 18231, "kind": "observation", "valid_at": "2026-10-19T09:00:00Z", "issued_at": null,
  "fetched_at": "2026-10-19T09:05:12Z", "measurement_basis": "model_analysis", "resolution_seconds": 3600,
  "location": { "id": 1, "slug": "chennai", "name": "Chennai" },
  "source": { "name": "open_meteo", "display_name": "Open-Meteo", "trust_level": "provider_model",
              "attribution_text": "<from API>" },
  "measurements": { "temperature_c": 31.4, "apparent_temperature_c": 36.0, "relative_humidity_pct": 68,
    "precipitation_mm": 0.0, "rain_mm": 0.0, "precipitation_probability_pct": null,
    "wind_speed_kmh": 14.2, "wind_gust_kmh": 27.0, "wind_direction_deg": 112 } }
```

PublicReport (StaffReport adds contact_email, evidence[].url, row_version, reviewed_by)

```
{ "id": "6f1c...-uuid", "event_type": "flood", "severity": "high", "description": "Water level rising on main road",
  "location_name": "T Nagar", "latitude": 13.0418, "longitude": 80.2341,
  "location": { "id": 1, "slug": "chennai", "name": "Chennai" }, "event_time": "2026-10-19T07:30:00Z",
  "evidence": [ { "type": "photo", "description": "Street view", "has_link": true } ],
  "status": "verified", "submitted_at": "2026-10-19T07:41:09Z", "reviewed_at": "2026-10-19T08:10:00Z" }
```

Timeseries

```
{ "metric": "precipitation_mm", "unit": "mm", "location": { "id": 1, "slug": "chennai", "name": "Chennai" },
  "kind": "observation", "interval": "day", "agg": "sum",
  "buckets": [ { "start": "2026-10-17T00:00:00Z", "value": 4.2, "sample_count": 24 },
               { "start": "2026-10-18T00:00:00Z", "value": null, "sample_count": 0 } ],
  "sources": ["open_meteo"], "trend": { "method": "ols", "slope_per_day": 0.8, "r_squared": 0.41, "n": 7 } }
```

Summary (pending and rejected appear only for staff; ingestion only for admin, else null)

```
{ "period": { "from": "2026-10-12T00:00:00Z", "to": "2026-10-19T00:00:00Z" },
  "weather": { "records_by_kind": { "observation": 1680, "forecast": 504 }, "locations_with_data": 6,
    "temperature_c": { "min": 22.1, "mean_of_location_means": 29.3, "max": 38.4 },
    "wind": { "mean_speed_kmh": 13.1, "max_gust_kmh": 51.0 } },
  "reports": { "total": 40, "by_status": { "verified": 22, "pending": 14, "rejected": 4 } },
  "sources": [ { "name": "open_meteo", "display_name": "Open-Meteo", "source_type": "weather_api",
                 "trust_level": "provider_model", "freshness": "fresh", "last_success_at": "2026-10-19T09:05:12Z" } ],
  "ingestion": { "runs_by_status": { "succeeded": 160, "partial": 3, "failed": 1 }, "success_rate": 0.976 } }
```

Anomaly (all four trailing fields are always present)

```
{ "valid_at": "2026-10-19T08:00:00Z", "metric": "temperature_c", "value": 41.2, "score": 4.1, "direction": "high",
  "baseline": { "median": 29.8, "mad": 1.9, "n": 720, "window_days": 30 }, "method": "modified_z_score",
  "threshold": 3.5, "explanation": "Value is 4.1 robust deviations above the 30-day median of 29.8 C, n=720",
  "is_official_warning": false, "verification": "unverified",
  "disclaimer": "Statistical flag only. Not an official warning." }
```

Run (admin) and **Source** (admin)

```
Run:    { "id": 311, "source": "open_meteo", "trigger": "manual", "status": "partial", "queued_at": "...Z",
          "started_at": "...Z", "finished_at": "...Z", "records_fetched": 504, "inserted": 120, "updated": 3,
          "skipped": 380, "failed": 1, "error_summary": "1 location failed: timeout", "details": [ ... ] }
Source: { "id": 1, "name": "open_meteo", "display_name": "Open-Meteo", "source_type": "weather_api",
          "trust_level": "provider_model", "is_enabled": true, "interval_seconds": 3600, "configured": true,
          "freshness": "fresh", "last_attempt_at": "...Z", "last_success_at": "...Z", "config": { } }
```

NewsItem and **Review**

```
NewsItem: { "id": 902, "title": "...", "summary": "...", "url": "https://...", "published_at": "...Z" or null,
            "category": "news", "is_official": false, "source": { "name": "...", "display_name": "...", "trust_level": "news
Review:   { "id": 55, "reviewer": { "id": "uuid", "display_name": "A. Rao" }, "from_status": "pending",
            "to_status": "verified", "reason_code": null, "private_notes": "Matched two photos", "created_at": "...Z" }
```

2.7 Report submission: request and validation

```
POST /api/v1/reports
{ "event_type": "flood", "severity": "high", "description": "Water level rising on main road",
  "location_name": "T Nagar", "latitude": 13.0418, "longitude": 80.2341, "event_time": "2026-10-19T07:30:00Z",
  "evidence": [ { "type": "photo", "url": "https://example.org/p1", "description": "Street view" } ],
  "contact_email": "name@example.com" }

201 { "id": "6f1c...", "status": "pending", "submitted_at": "...Z", "tracking_token": "<shown once>" }
```

Field	Rule
event_type	Required, enum value
severity	Optional enum
description	Required, 10 to 2000 characters
location_name	Optional, max 200 characters
latitude, longitude	Required, -90 to 90 and -180 to 180
event_time	Required, ISO 8601 with timezone; not more than 5 minutes in the future; not older than 30 days
evidence	Optional, max 5 items; type photo, video, link or other; url must be https and at most 2048 characters; description max 200. Metadata only, no file upload
contact_email	Optional, valid email, never displayed publicly
Other fields	Unknown fields (including status) are rejected with 422

2.8 What each audience sees

Data	Anonymous	Reviewer	Admin
Weather, locations, news	Yes	Yes	Yes
Verified reports (public fields, no evidence URLs)	Yes	Yes (full)	Yes (full)
Pending and rejected reports, contact email, evidence URLs	No (own report status via tracking token)	Yes	Yes
Summary: pending and rejected counts	No	Yes	Yes
Review history and private notes	No	Yes	Yes
Ingestion runs, run trigger, sources	No	No	Yes

2.9 Integration notes

- Generate TypeScript types from `/openapi.json` (for example with openapi-typescript) once the backend is running, and treat any difference from this document as a bug to raise with the backend owner.
- Fields may be added without notice (ignore unknown fields); removals or renames are breaking changes and require a version bump agreed with you in advance.
- Rate limits are per IP and in-memory in the MVP; do not retry aggressively on 429.
- Ingestion runs only when the backend worker is running and a source is enabled; an empty dashboard right after setup is expected, not a frontend bug.
- Open items to confirm with the backend owner: final list of seeded locations, whether the anomalies screen ships in the MVP, and the production API and frontend origins.